

Article

Analysis of CNN-Based Deep Learning Architectures and Performance Enhancement Strategies for Pneumonia Classification Using Chest X-Ray Images

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Abstract

Background: Deep learning models, particularly convolutional neural networks (CNNs), have shown promising performance for pneumonia detection using chest X-ray images. However, the impact of preprocessing, architecture selection, data augmentation, and ensemble strategies has not been systematically evaluated. This study investigated how these factors affect model robustness and diagnostic performance. **Methods:** A public pediatric chest X-ray dataset was used to systematically evaluate pixel normalization methods, six CNN architectures, progressive data augmentation strategies for class imbalance, and both feature-level and decision-level ensemble approaches. Model performance was assessed by considering not only overall classification accuracy but also clinically relevant risk metrics, particularly false-negative rates. **Results:** Pixel normalization to the 0–1 range improved model convergence, while Xception and InceptionV3 achieved the best overall performance. Model-specific augmentation strategies were more effective than a fixed 1:1 class ratio for reducing false negatives. Feature-level ensembles tended to overfit, whereas decision-level ensembles provided more stable but only modest performance improvements. **Conclusions:** These findings demonstrate that reliable medical AI systems require systematic optimization of preprocessing techniques, model architecture, data augmentation strategies, and clinically meaningful evaluation metrics rather than maximizing a single performance indicator. The proposed framework provides practical guidelines for developing robust deep learning models for pneumonia diagnosis in clinical settings.

Keywords: deep learning; convolutional neural network (CNN); pneumonia; chest X-ray; transfer learning



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1. Introduction

Deep learning is a branch of machine learning that uses multiple nonlinear transformations to extract and summarize high-dimensional information from large datasets, thereby incorporating aspects of human cognitive processing [1]. Its rapid development has been supported by advances in GPU-based parallel computing and the availability of large-scale public datasets such as ImageNet [2].

Convolutional neural networks (CNNs) are a type of deep learning architecture that learns hierarchical visual features through convolution and pooling. Following the success of AlexNet, CNNs became a mainstream approach to image classification, with subsequent architectures further improving recognition performance [2]. These advances have spread

across diverse fields and show particular promise in medical imaging, where large volumes of visual data require precise analysis [1].

Deep learning has shifted medical image interpretation from experience-dependent qualitative judgment to data-driven quantitative analysis [3]. Although CT, MRI, X-ray, and ultrasound images are essential for diagnosis, treatment planning, and prognosis prediction, their interpretation requires expertise and may vary among evaluators. Deep learning has been proposed to reduce this variability and has achieved performance comparable to or exceeding that of human experts in several clinical fields [3,4].

Pneumonia is a major respiratory infectious disease associated with substantial morbidity and mortality, particularly among children and older adults [5]. It causes alveolar inflammation and exudate accumulation, which can rapidly impair respiratory function and progress to respiratory failure or sepsis. Therefore, early diagnosis and prompt treatment are critical determinants of patient prognosis [6].

Chest X-ray is a standard imaging method for pneumonia diagnosis, but overlapping anatomical structures and the two-dimensional projection of three-dimensional tissues can obscure lesions. Interpretation also depends on specialist experience, image quality, and patient condition, which may reduce diagnostic consistency [7]. These limitations motivate the use of AI-based methods for automated and quantitative image analysis.

As shown in the X-ray images above, images of normal cases display clear lung fields, whereas those of bacterial pneumonia cases tend to show localized infiltrates with well-defined borders, and those of viral pneumonia cases tend to show diffuse ground-glass opacities (GGOs) with indistinct borders (Figure 1). However, in actual clinical practice, there is a limitation in that the lesions often appear subtle or atypical, making objective differentiation difficult.

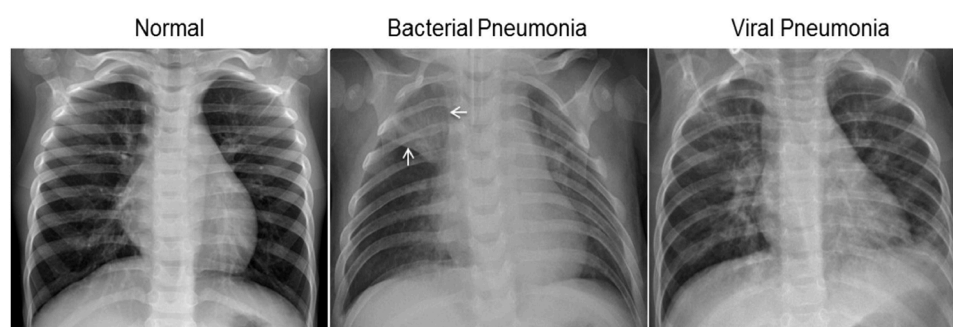


Figure 1. Examples of chest X-ray images [8].

To overcome these diagnostic limitations, studies applying artificial intelligence, particularly deep learning, to medical image analysis are currently underway. In fact, deep learning models such as ResNet and Swin Transformer have demonstrated a high accuracy of over 98% in pneumonia classification on chest X-ray images, indicating their technical feasibility [7]. However, previous studies tend to focus on presenting results regarding the final performance of specific models. As a result, there is a relative lack of systematic and quantitative comparative analyses regarding how various model architectures, hyperparameters, and training techniques specifically affect actual performance (accuracy, training speed, generalization performance, etc.).

This study quantitatively evaluates how major modeling decisions affect CNN-based binary classification of normal and pneumonia chest X-ray images. Using a public dataset, we systematically compare pixel normalization, six pretrained CNN architectures, fixed learning rates, and feature- and decision-level fusion. Rather than proposing a new architecture, this study aims to identify model-specific trade-offs in convergence behavior

and ensemble performance, thereby providing practical guidance for the development of pneumonia diagnostic-support models.

2. Literature Review

Pneumonia classification research using chest X-ray images has evolved from the application of single CNN models toward imageNet-based transfer learning, multi-architecture comparison, data augmentation, class imbalance mitigation, and ensemble learning. Recently, the scope of research has been expanded beyond the binary classification that distinguishes normal from pneumonia to include analysis of pneumonia type, lesion location, and severity and to support clinical decision-making. However, it is difficult to directly compare performance, as previous studies use different datasets, data segmentation techniques, input resolutions, learning ranges, and evaluation indicators in pre-learning models. Therefore, in this section, data composition, transfer learning structure, data augmentation and class imbalance processing, ensemble methods, and verification strategies are compared to examine chest X-ray-based pneumonia classification, and the strengths and limitations of each approach are analyzed.

Kermamy et al. [8] proposed an image-based deep learning framework and suggested that transfer learning can be used to classify normal and pneumonia cases in pediatric chest X-ray images. This study showed that, by transferring pre-trained features from large-scale general images to medical image analysis, a diagnostic model can be developed, even with relatively limited training data in the medical field. Kermamy et al.'s research is significant because it provided the basis for the field of childhood pneumonia classification. However, as the main purpose of the study was to demonstrate the diagnostic possibility of image-based deep learning, the effects of different CNN structures, preprocessing strategies, learning rates, data augmentation techniques, and ensemble methods were not analyzed separately under the same conditions.

Chouhan et al. [9] applied an ensemble technique combining five pre-trained CNN models to the classification of pediatric chest X-ray pneumonia, achieving an improved accuracy of 96.4% and a reproduction rate of 99.62% compared to a single model. The strength of this study is that, instead of using only one CNN, it utilized the complementarity of features extracted by different pre-trained models and incorporated transfer learning, data augmentation, and ensembles into one pneumonia classification process. However, as the study focused on improving the performance of the final ensemble, it is difficult to determine whether the response of each model differed depending on the level of data augmentation and to what extent each preprocessing technique, learning rate, and ensemble contributed to the change in performance. In addition, although high reproduction rates were reported, an analysis of the clinical risk by comparing the changes in false negatives and false positives according to the level of augmentation was not included in the main study scope.

Kundu et al. [10] developed an automatic pneumonia detection system based on GoogLeNet, ResNet-18, and DenseNet-121. In addition, an ensemble technique was proposed to determine weights by combining four performance indicators (precision, reproducibility, F1-score, and AUC). In particular, the study objectively showed that the system demonstrated improved performance compared to a single model in terms of robustness and reliability, achieving 98.81% and 86.85% accuracy through 5-fold cross-validation and statistical verification on multiple datasets (Kermamy and RSNA, respectively). However, the study is limited in that the analysis focused on decision-level fusion without conducting a direct comparison with feature-level fusion, and the effects of individual learning conditions were not analyzed separately.

Wang et al. [7] developed a fully automated deep learning pipeline that incorporates the standardization of chest X-ray imaging, lesion visualization, and disease diagnosis. The system was developed with 145,202 heterogeneous, multi-center images, and retrospective and prospective evaluations were performed with thousands of additional images obtained from multiple countries and four patient cohorts. The system showed an AUC ranging from 0.94 to 0.98 on a pneumonia-associated classification task, and an independent 440-page chest X-ray evaluation showed similar performance to that of a senior radiologist; thus, it may improve the interpretation performance of junior radiologists. However, as this study aimed to establish an integrated diagnostic system that included pneumonia type and severity, it differs in purpose and scope from studies aiming to control and compare the individual effects of CNN structures, normalization methods, learning rates, augmented ratios, and fusion levels on a specific pediatric binary classification dataset.

In addition, Hashmi et al. [11] supplemented the limitations of using a single model by fine-tuning ResNet18, Xception, InceptionV3, DenseNet121, and MobileNetV3 and weight-combining the predictions of each model. However, as the analysis focused on improving the performance of the final weighted classifier, the effects of the augmented sensitivity or individual learning conditions of each model were not clearly separated. Ravi et al. [12] went beyond this performance-oriented approach to apply cost-sensitive learning, transfer learning-based feature fusion, and stacked ensemble reflecting class imbalances and misclassification costs. This study is significant in that it considered class imbalance and feature-level fusion together, but it did not compare the performance and overfitting patterns of each model while gradually changing the amount of augmentation. Table 1 summarizes the data compositions, CNN architectures, imbalance processing methods, ensemble methods, major strengths, and methodological limitations applied in the abovementioned studies for chest X-ray pneumonia classification.

Table 1. Comparison of representative studies of chest X-ray-based pneumonia classification.

Parameter	Approach	Contribution	Limitation
Kermany et al. [8]	Classification based on transfer learning	Presents pediatric pneumonia data and image-based transfer learning	Does not compare controls according to design elements
Chouhan et al. [9]	Multiple pre-trained CNNs and ensembles	Improves performance with complementarity across CNNs	Difficulty in determining the individual contributions of augmentation methods, models, and ensembles
Kundu et al. [10]	Metric-based weighted ensemble	Assesses robustness with multiple datasets and cross-validation	Focuses on decision-level fusion
Wang et al. [7]	Integrated diagnostic pipeline	Compares external validation and clinical assessment	Does not compare individual design elements
Hashmi et al. [11]	Multiple CNNs and weighted classifier	Combines predictions from diverse architectures	Does not analyze the impact of model-specific training strategies
Ravi et al. [12]	Cost-sensitive learning and feature fusion	Accounts for imbalance and misclassification costs	Does not compare responses according to augmentation level or model

In summary, previous studies have demonstrated that transfer learning, data augmentation, class imbalance handling, and ensemble techniques can improve the performance of chest X-ray-based pneumonia classification. However, because data partitioning, input resolution, backbone freezing or fine-tuning ranges, augmentation methods, and evaluation metrics vary across studies, it is difficult to clearly determine whether reported performance differences stem from the CNN architecture itself or from different training conditions. Furthermore, as many studies implemented the architecture, augmentation process, and ensemble techniques at the same time and evaluated performance primarily based on the

final accuracy or AUC, there are limitations in determining the effects of each design element and the responses of individual models. In particular, studies that stepwise compare class-specific performance changes according to augmentation levels with clinically significant errors (such as false negatives, where actual pneumonia patients are classified as normal) are limited. To address these limitations, this study sequentially evaluates pixel normalization, six ImageNet pre-trained CNN architectures, sensitivity to a fixed learning rate, and feature- and decision-level fusion using identical data and typical experimental conditions.

3. Materials and Methods

3.1. Dataset

This study utilized the “Chest X-Ray Images (Pneumonia)” dataset published by Kermany et al. (2018) [8]. The dataset consists of 5863 anteroposterior and lateral chest X-ray images of pediatric patients collected at Guangzhou Medical Center. The data is publicly available on Mendeley Data, its original repository; for this study, it was collected via the Kaggle platform and used for analysis.

The images in this dataset were obtained from pediatric patients aged 1 to 5 years, and all of them are pre-classified into two categories (normal and pneumonia). According to the original paper, the dataset underwent a quality control process to remove low-quality or unreadable images before being used for training in order to improve reliability. Two specialists then interpreted and classified the images through cross-validation, and a third expert finally re-examined the evaluation set to ensure the accuracy of the labels [8].

The dataset provided on the Kaggle platform was pre-split into training, validation, and test subsets containing 5216, 16, and 624 images, respectively. To address the limited size of the original validation subset (16 images), which was insufficient to provide a statistically stable estimate of model performance during development, the original training and validation subsets were pooled (5232 images) for re-splitting. Prior to re-splitting, images sharing a patient identifier (extracted from the filename for the pneumonia class), an exact duplicate (SHA-256 hash match), or a near-duplicate (perceptual hash, Hamming distance ≤ 5) were grouped. Any group found to overlap with the fixed test set was excluded from the pool in its entirety. Using mutually exclusive group-level exclusion categories, 468 images were excluded from groups identified by patient-identifier overlap, no additional images were excluded under the exact-duplicate-only category, and 102 images were excluded from groups identified by near-duplicate matching. In total, 570 of the 5232 pooled images were excluded, leaving 4662 images for re-splitting. The remaining pool ($n = 4662$) was then re-split at the group level, rather than the image level, into a new training set ($n = 3967$) and a new validation set ($n = 695$) using stratified random sampling by class (85:15 ratio, fixed seed = 123), ensuring that all images belonging to the same patient or duplicate group remained within a single subset (Figure 2). File-level and, for the pneumonia class, patient-level disjointness between the training and validation sets were subsequently verified to confirm that no leakage was introduced by the re-splitting procedure.

This new validation set was used for all performance-based model development decisions, including early stopping, architecture comparison, learning-rate selection, and ensemble-weight selection. During the revised data-preparation procedure, the original 624-image test subset was accessed only to identify patient-level and image-level overlap with the pooled training and validation data. Under the revised protocol, the test subset was not used to compare candidate models, tune hyperparameters, select learning rates, or determine the ensemble configuration. After all performance-based model development decisions had been finalized using the new validation set, the test subset was used to report the final performance of the selected model.

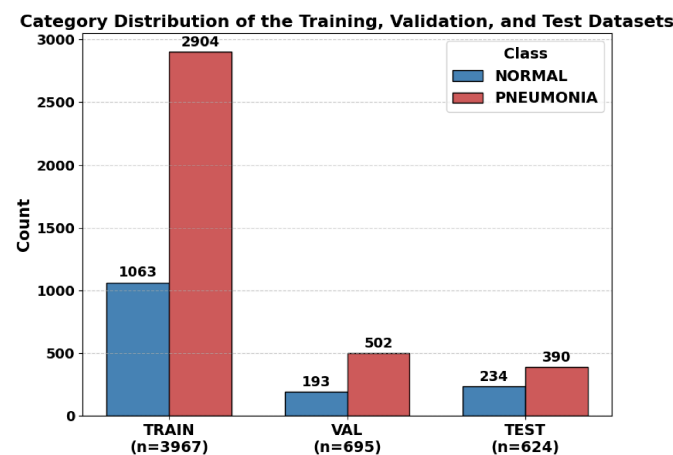


Figure 2. Class distribution of the training, validation, and test datasets.

3.2. Experimental Design

In this study, experiments were conducted under controlled conditions with identical hyperparameter settings in order to fairly and objectively compare performance differences resulting from the structural characteristics of various transfer learning models.

To ensure a controlled comparison across the six architectures, all chest X-ray images were resized to a common resolution of 256×256 pixels. This resolution was selected as a practical compromise between preserving spatial information and controlling computational and memory requirements. Applying the same input resolution to all models also prevented differences in image dimensions from becoming an additional confounding factor in the architecture comparison. Therefore, “ 256×256 pixels” was used as a standardized experimental condition rather than as an empirically optimized resolution. In addition, the batch size was set to a standard value of 32, considering the stability of learning and generalization performance together.

We adopted Adam, a widely used optimizer in the field of deep learning. In addition, we applied the early stopping technique to prevent overfitting during the training process and to improve the model’s generalization performance. Early stopping is a method that automatically halts training if validation loss does not improve for a certain number of epochs; it helps to prevent unnecessary training and contributes to obtaining optimal weights. By maintaining these consistent settings, we ensured the reliability and reproducibility of model comparisons. Table 2 summarizes the experimental environment and hyperparameter configurations used in this study.

Table 2. Experimental design for transfer learning.

Parameter	Configuration
Input Image Shape	$256 \times 256 \times 3$
Batch Size	32
Pooling Layer	Global Average Pooling
Classifier	Dense Layer $\times 1$
Output Activation	Sigmoid
Optimizer	Adam
Loss Function	Binary Cross Entropy
Metrics	Accuracy
Early Stopping	Based on val_loss, Patience = 5
Optimal Weight Restoration	Restore Best Weight = True

3.3. Image Preprocessing and Pixel Value Normalization

The performance of deep learning models is greatly influenced by the quality and consistency of the training data, thus making the data preprocessing step crucial [13]. In particular, the distribution of pixel values in medical images can vary slightly depending on the imaging environment or equipment, which can destabilize the model's training [14]. Therefore, in the first experiment, we applied pixel value normalization to improve the training stability of the deep learning model. In this study, we applied 0–1 scaling (min–max scaling), one of the most widely used normalization techniques.

This method involves dividing the integer value (between 0 and 255) of each pixel in an image by 255.0 in order to convert it into a real number between 0.0 and 1.0. This scaling adjustment plays a key role in helping the model update its weights more stably during training and converge to optimal performance at a faster rate.

Figure 3 compares the pixel value distributions before and after applying 0–1 scaling to the image data used in this study. Before normalization, pixel values are distributed over a wide range from 0 to 255, whereas after normalization, all values are rescaled to a range of 0–1. Although the overall shape of the distribution remains largely unchanged because normalization is a linear scaling process, the numerical range of the input values is standardized without altering the intrinsic structural information of the chest X-ray images. This standardization facilitates more stable gradient optimization, improves convergence during training, and ensures consistent input conditions across all CNN architectures while preserving the statistical characteristics of the original images.

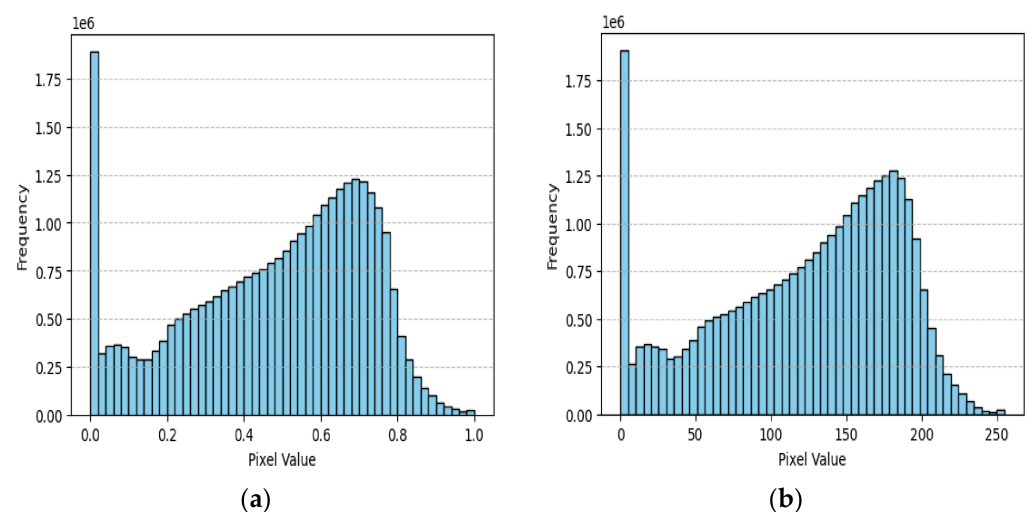


Figure 3. Comparison of pixel value distributions before and after normalization: (a) Pixel value distribution before normalization. (b) Pixel value distribution after normalization.

3.4. Transfer Learning Model Architectures

For this study on binary classification, we selected a total of six convolutional neural network architectures as comparative transfer learning models: VGG16, ResNet, Inception V3, Xception, EfficientNet B0, and DenseNet 121. This family of models utilized standard benchmark models widely used to ensure the reliability and reproducibility of experimental results. At the same time, the models were designed in a variety of configurations to enable a systematic analysis of the multifaceted trade-offs between architectural complexity, computational cost, and diagnostic performance. The key features of each model are as follows: VGG16 is characterized by a uniform architecture that achieves network depth by repeatedly stacking small 3×3 convolutional filters. It is a standard baseline model

widely used in image classification research due to its simple yet powerful performance. It provides a basic benchmark for comparing performance with complex models [15].

ResNet is an innovative model that introduced the concept of residual learning, enabling the training of very deep networks; because of its ability to effectively extract fine-grained, hierarchical features from complex medical images, it has established itself as a core architecture in chest X-ray classification research [16].

Inception V3 achieves both multi-scale feature extraction and computational efficiency by utilizing the Inception module, which processes filters of various sizes (1×1 , 3×3 , 5×5 , etc.) in parallel within a single layer; it has demonstrated high potential in the field of pneumonia diagnosis through studies comparing its performance with that of other standard models [17].

Xception is a model that significantly improves parameter efficiency by extensively utilizing depthwise separable convolutions, which maximize the concepts of Inception; it has demonstrated high performance in various studies in classifying chest diseases, including COVID-19, and it is regarded as a modern, high-performance baseline model [18].

EfficientNet B0 is a model designed to achieve high accuracy with a small number of parameters; it has set a new standard for computational efficiency through a compound scaling technique that balances the expansion of the network's depth, width, and resolution. This is a critical consideration when deploying the model in real-world medical diagnostic systems with limited resources [19].

Finally, DenseNet 121 features an innovative architecture that maximizes feature reuse and mitigates the vanishing gradient problem through dense connections between all layers. In particular, as it served as the core model in CheXNet—a landmark study in the field of chest X-ray image analysis—and has proven its performance, it is an indispensable benchmark in credible research in this field [20].

In this study, all six architectures were initialized with ImageNet-pretrained weights and used without their original classification layers (`include_top = False`). Global average pooling was applied to the output of each convolutional backbone, followed by a single dense layer with sigmoid activation for binary classification. All layers of the pretrained backbones were frozen throughout the architecture comparison, learning-rate evaluation, and ensemble experiments; therefore, no backbone fine-tuning was performed, and only the newly added classification layer was updated during training. The number of trainable parameters consequently depended on the dimension of the pooled feature vector, as summarized in Table 3.

Table 3. Transfer learning configuration and trainable parameters.

Architecture	Pooled Feature Dimension	Total Trainable Parameters
VGG16	512	513
ResNet50	2048	2049
InceptionV3	2048	2049
Xception	2048	2049
EfficientNetB0	1280	1281
DenseNet121	1024	1025

3.5. Learning Rate Sensitivity Analysis

The learning rate is one of the most influential hyperparameters in deep learning because it controls the step size used to update model weights during gradient-based optimization. Changes in the learning rate can affect not only predictive performance but also convergence speed, loss stability, and the overall learning trajectory. Therefore, this

experiment focused on examining the relationship between learning rate magnitude and model behavior rather than identifying a single globally optimal value [21].

Five fixed learning rates from 1.0×10^{-5} to 1.0×10^{-1} were examined at logarithmic intervals to explore their relationships with accuracy, loss, convergence speed, and training stability. Each value remained constant throughout its training run, thus allowing a direct comparison of learning behavior under distinct rate conditions. The experiment was therefore an exploratory sensitivity analysis rather than an exhaustive grid search. Future work may combine this analysis with grid search, learning rate finders, or dynamic scheduling to refine the range and adapt the learning rate during training.

3.6. Ensemble Model Design

The predictive performance of a single deep learning model tends to be constrained by the inherent limitations of the dataset or architecture used. To overcome these limitations, ensemble techniques—which combine multiple models to derive a final prediction—are proposed as an effective alternative. The core of ensemble learning lies in compensating for the bias or variance that individual models may possess, thereby yielding results that are more stable and exhibit superior generalization performance [22].

In ensemble techniques, there are two main approaches—feature-level fusion and decision-level fusion—which differ based on the stage at which information is fused [23]. In this study, we aim to compare and analyze these two techniques in a controlled environment to observe the potential benefits of each approach.

Feature-level fusion, also known as early fusion, is a method in which information from multiple models is integrated prior to the prediction stage, specifically during the feature extraction stage [23]. In the feature-level fusion model used in this study, models with different architectures, such as Inception V3 and Xception, each act as independent feature extractors. Each model generates unique feature maps or feature vectors from the input image, and these individually extracted features are combined into a single, integrated fusion feature representation through methods such as concatenation.

Finally, this rich, high-dimensional feature representation is fed into a single classifier to perform the final prediction. The fundamental philosophy of this approach is to enable the model to simultaneously utilize all information extracted from different perspectives from the early stages of training, thereby learning the complex interactions among features [23]. This overall process is illustrated in Figure 4.

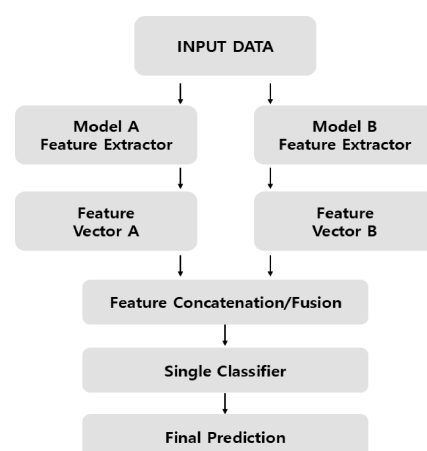


Figure 4. Structure of the feature-level fusion model.

Late fusion is a fusion method in which each model undergoes complete training and makes predictions independently, and the results are then combined. In the final stage of

the ensemble, these independent prediction results are aggregated, and a final decision is made using aggregation rules such as soft voting or weighted averaging. This process is analogous to treating each model as an independent expert and deriving the most reliable conclusion based on their combined opinions (Figure 5) [23].

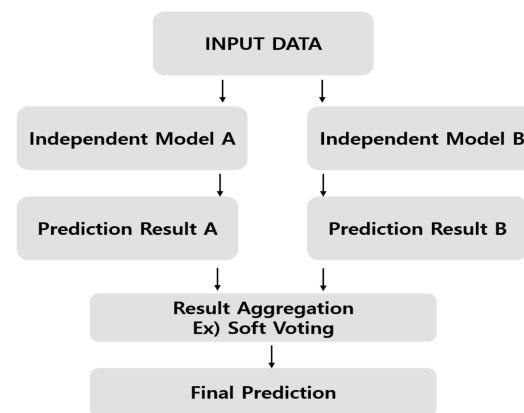


Figure 5. Structure of a multi-level decision fusion model.

In the multi-level fusion model used in this study, the pretrained Xception and InceptionV3 backbones were frozen, and a new output layer was attached to and trained for each, producing a logit for the same input data. Subsequently, a weighted average method was adopted to integrate the logits from the individual branches prior to a final sigmoid activation. The ensemble weights were selected using only the new validation set, without reference to test-set performance. In the preceding validation experiments, Xception showed more stable performance across the examined learning-rate range, whereas InceptionV3 exhibited greater degradation at the highest learning rate. Therefore, a slightly larger contribution was assigned to Xception while retaining a substantial contribution from InceptionV3. The fused logit was calculated as 0.6 times the Xception logit plus 0.4 times the InceptionV3 logit, followed by sigmoid activation. This weighting scheme was heuristically selected as a model-specific design choice based on the validation behavior observed in the present experiments, rather than as a universally optimal weighting applicable to other models or datasets.

4. Results

4.1. Analysis of the Effects of Pixel Value Normalization

Figure 6 compares the validation-set training trajectories of the Xception model with and without pixel normalization. The most visually apparent difference is in training stability: the unnormalized configuration exhibited noticeably noisier, more fluctuating validation curves and triggered early stopping after approximately 24 epochs, whereas the normalized configuration converged along a markedly smoother trajectory and continued training for the full 50 epochs. Over the shared training range, the normalized configuration maintained higher validation accuracy and lower validation loss than the unnormalized configuration, with no crossover at any point (final validation accuracy: 0.9165 vs. 0.9468; final validation loss: 0.2508 vs. 0.1230). As this comparison is based on a single representative training run, these values should be interpreted as illustrative of the qualitative benefit of normalization rather than as a precisely quantified effect size.

This pattern is consistent with the expected role of pixel normalization: by constraining input values to a common $[0, 1]$ range, normalization reduces scale-driven variance in gradient updates, allowing the network to learn the relative differences between pixel intensities—such as the boundaries, shading, and patterns of pneumonia lesions—more

precisely [14]. Accordingly, pixel normalization is best understood as a stabilizing pre-processing step that improves training reliability, rather than a decisive factor in absolute classification performance.

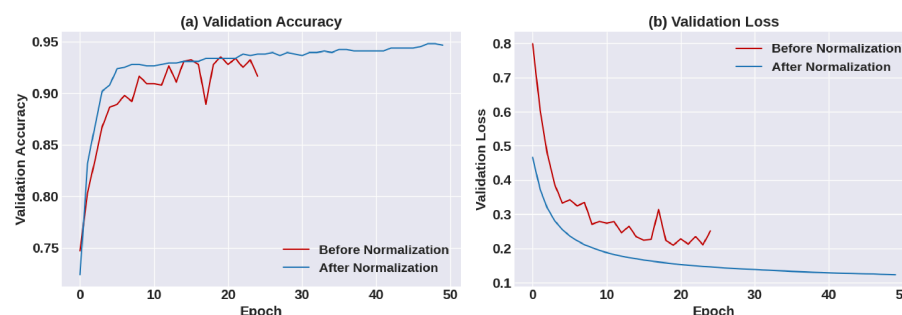


Figure 6. Comparison of validation performance for the Xception model with and without pixel normalization, overlaid across training epochs: (a) validation accuracy; (b) validation loss.

4.2. Performance by Transfer Learning Model

To account for run-to-run variability in the random initialization of the classification head, each of the six architectures was trained independently five times using different random seeds, and performance is reported as the mean $\pm$ standard deviation across runs (Table 4). InceptionV3, DenseNet121, and Xception achieved the highest mean validation accuracies (0.947–0.951) and substantially lower losses than the remaining models. VGG16 showed intermediate performance with a mean validation accuracy of 0.892, whereas ResNet50 and EfficientNetB0 achieved lower mean accuracies of 0.765 and 0.722, respectively. The differences in mean accuracy among the three top-performing models were relatively small. Therefore, these results are interpreted descriptively, and no claim of statistical equivalence or significant pairwise differences is made.

Table 4. Mean classification performance of six frozen pretrained CNN architectures.

Model	Accuracy (Mean $\pm$ Std)	Loss (Mean $\pm$ Std)
InceptionV3	0.951 $\pm$ 0.001	0.133 $\pm$ 0.001
DenseNet121	0.949 $\pm$ 0.004	0.132 $\pm$ 0.002
Xception	0.947 $\pm$ 0.002	0.124 $\pm$ 0.001
VGG16	0.892 $\pm$ 0.002	0.285 $\pm$ 0.002
ResNet50	0.765 $\pm$ 0.002	0.440 $\pm$ 0.001
EfficientNetB0	0.722 $\pm$ 0.000	0.591 $\pm$ 0.000

This performance difference is attributed to the structural characteristics of each model. As previously explained, Inception V3 is based on the Inception module, which applies filters of various sizes in parallel within a single layer [22]. This architecture is highly effective for multi-scale feature extraction, meaning that it can simultaneously capture both large-scale features—such as the diffuse ground-glass opacities seen in pneumonia X-ray images—and fine-scale features, such as localized pulmonary infiltrates.

Xception, which stands for “Extreme Inception”, is a model that maximizes the concepts of Inception [18]. Based on the assumption that inter-channel and spatial correlations can be learned completely independently, in Xception, the Inception modules are replaced with depth-wise separable convolutions.

This approach uses parameters much more efficiently to enhance the model’s learning ability, helping to prevent the overfitting that can occur with limited medical datasets

and improve the model's generalization performance. Based on this top-tier separation, Xception and Inception V3 were selected—considering both architectural structure and extensibility to the subsequent optimization stages—to observe how performance evolves through further experiments, while DenseNet121, structurally distinct in its feature-reuse design, was reserved for future work, as discussed in Section 5.

4.3. Effects of Learning Rate on Model Performance

Figure 7 reveals that both models followed a similar trend at the lower end of the learning rate range but diverged in their sensitivity to excessively high learning rates. At a learning rate of 1.0×10^{-5} , learning proceeded slowly for both models, yielding below-optimal validation accuracy (Inception V3: 0.899; Xception: 0.850) rather than a complete failure to learn. At the upper end, however, the two architectures behaved differently: Inception V3 showed a clear overshooting pattern at a learning rate of 1.0×10^{-1} , with validation loss rising sharply and accuracy dropping to 0.909, whereas Xception remained comparatively robust at the same learning rate, with accuracy declining only marginally (0.962) despite an increase in validation loss. This suggests that Xception's depthwise separable convolutions may confer greater tolerance to large weight updates than Inception V3's parallel multi-scale filtering.

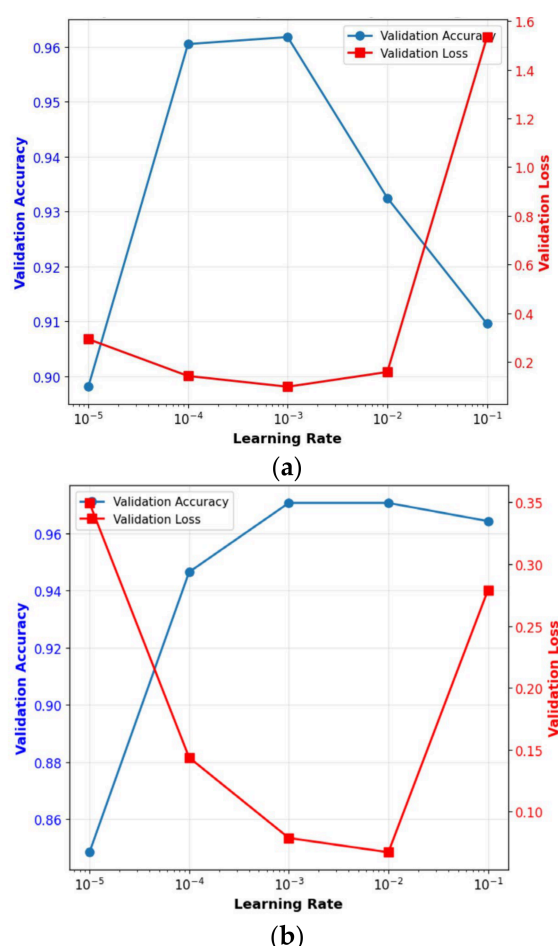


Figure 7. Changes in accuracy and loss values by learning rate: (a) Inception. (b) Xception.

In terms of quantitative performance, the Inception V3 model achieved comparable peak validation accuracies of 0.961 and 0.962 at learning rates of 1.0×10^{-4} and 1.0×10^{-3} , respectively. The Xception model reached its peak accuracy of approximately 0.965 at

learning rates of 1.0×10^{-3} and 1.0×10^{-2} , and it maintained a high accuracy of 0.962 even at 1.0×10^{-1} , demonstrating stable performance over a wider range than Inception V3.

The learning rate is a parameter that determines not only the final quantitative value but also the convergence process itself. Therefore, we determined that it was necessary to compare not only scores but also learning efficiency, and, thus, the convergence speed and stability of each learning process were visualized.

Figure 8 shows the results of an analysis of convergence speed and stability for different learning rates based on changes in batch-wise loss. For Inception V3, the loss reached a 50% reduction within a comparable number of batches at 1.0×10^{-2} and 1.0×10^{-3} (46 and 48 batches, respectively), with the latter showing a smoother decline, representing the best balance between convergence speed and stability; at 1.0×10^{-5} , the loss failed to reach a 50% reduction within the training run, indicating that learning had nearly stalled. For Xception, although there was some initial fluctuation at a learning rate of 1.0×10^{-2} , it reached the 50% loss reduction point the fastest among all tested rates (18 batches), showing high training efficiency, whereas at 1.0×10^{-5} convergence was markedly slower, requiring 2225 batches to reach the same threshold.

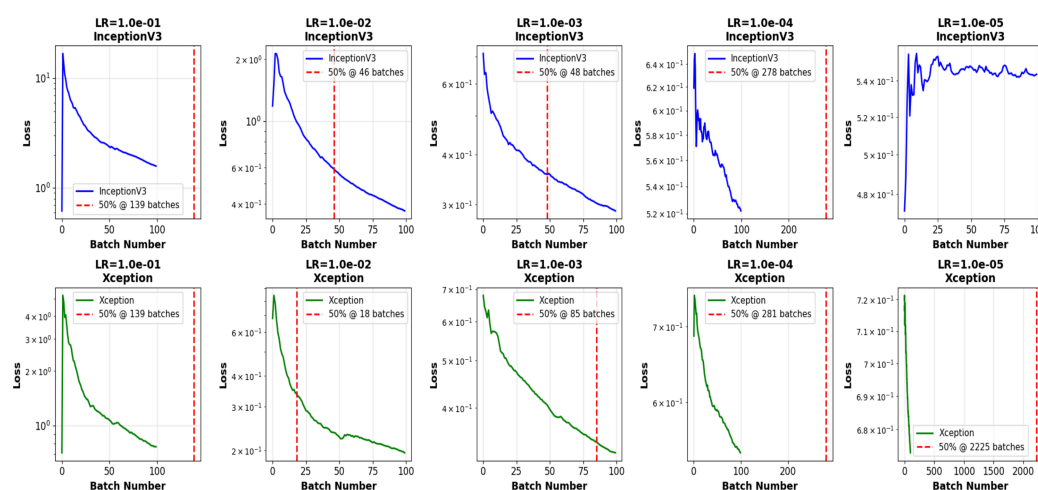


Figure 8. Training loss convergence of the Inception V3 (top row) and Xception (bottom row) models for five different learning rates.

Overall, the results demonstrated that learning rate magnitude affected the two architectures differently in terms of predictive performance, convergence speed, and loss stability. Based on these observed relationships, 1.0×10^{-3} for InceptionV3 and 1.0×10^{-2} for Xception were adopted as representative stable settings for the subsequent ensemble experiments. These values were selected to maintain consistent experimental conditions and should not be interpreted as globally optimal learning rates.

4.4. Analysis of Model Ensemble Effects

Figure 9 compares the validation accuracy and validation loss trajectories of the Xception and InceptionV3 single models together with the feature-level fusion model over training. Consistent with Xception's faster convergence behavior reported in Section 4.3, both Xception and the fusion model triggered early stopping after approximately 11 epochs, whereas InceptionV3 continued training for approximately 35 epochs before its validation performance plateaued.

Among the three, the feature-level fusion model achieved the highest validation accuracy in the earliest epochs, but this early advantage narrowed over the course of training. By the final epoch, Xception attained both the highest validation accuracy (0.9669) and the lowest validation loss (0.0983), InceptionV3 reached a lower validation accuracy (0.9540)

and a higher validation loss (0.1206), and the fusion model settled at an intermediate validation accuracy (0.9626) but the highest validation loss of the three (0.1309). This pattern is consistent with the increased dimensionality of the fusion model's combined feature space: concatenating features from both backbones allowed the model to fit the training distribution quickly and reach a strong validation accuracy early on, but also left it more susceptible to overfitting as training progressed. In contrast, Xception's more compact single-model representation converged to a comparatively lower and more stable validation loss over the same training range.

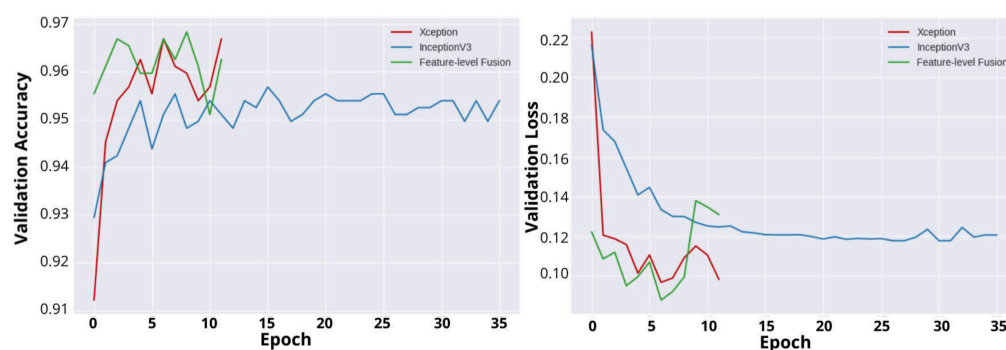


Figure 9. Comparison of validation accuracy and validation loss between the Xception and InceptionV3 single models and the feature-level fusion model, overlaid across training epochs.

Figure 10 shows the validation accuracy and validation loss of the decision-level fusion model over training. Its validation accuracy started lower than that of all three models discussed above, but the model's learning progressed with markedly less epoch-to-epoch fluctuation than either Xception or the feature-level fusion model, settling into a smooth, steadily improving trajectory.

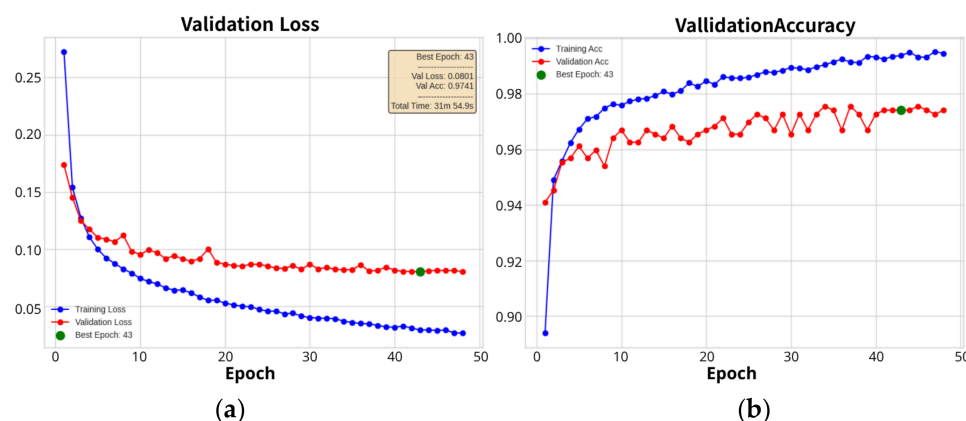


Figure 10. Training and validation accuracy and loss of the decision-level fusion model by epoch, with the best epoch (43) marked: (a) loss; (b) accuracy.

By the best epoch (epoch 43 of 48), the decision-level fusion model reached a validation accuracy of 0.9741 and a validation loss of 0.0801, the highest validation accuracy and lowest validation loss among all four models compared in this study. This indicates that, unlike feature-level fusion, combining the independent predictions of the two backbones after training—rather than combining their internal feature representations—avoided the overfitting associated with high-dimensional feature concatenation while still yielding a measurable performance gain over either single model.

4.5. Final Model Performance on the Test Set

To evaluate the performance of the finally selected design, the decision-level fusion model combining the frozen Xception and InceptionV3 backbones (Section 3.6) was evaluated on the fixed test set ($n = 624$). During the revised protocol, this test set had been accessed for overlap screening during data preparation but was not used for performance-based model selection or hyperparameter tuning. On this test set, the model achieved an accuracy of 0.8109, an AUC of 0.9156, and a loss of 0.7230 (Figure 11).

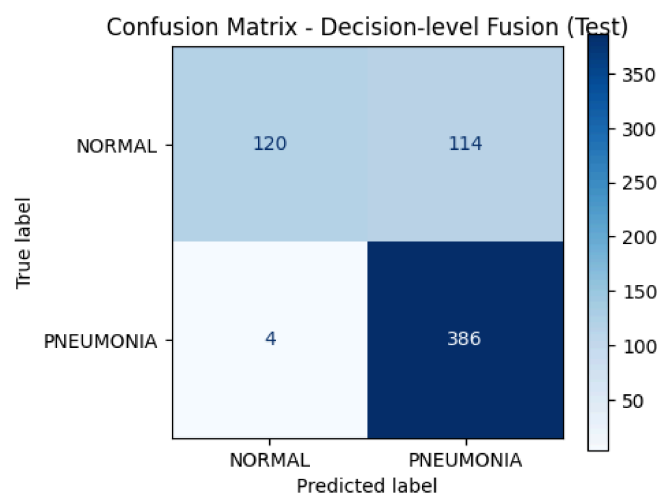


Figure 11. Confusion matrix of the decision-level fusion model on the fixed test set.

The confusion matrix (Figure 11) revealed a pronounced class-wise asymmetry that had not been apparent from the validation-set accuracy and AUC alone: only 4 of 390 pneumonia cases were misclassified as normal (99.0% recall), whereas 114 of 234 normal cases were misclassified as pneumonia (51.3% recall). This asymmetry is consistent with two characteristics of the experimental design. First, the training set was more severely imbalanced toward the pneumonia class (73.2% pneumonia) than the test set (62.5% pneumonia), so a model calibrated to this training-set imbalance would be expected to over-predict pneumonia on a test set containing proportionally more normal cases. Second, because the experimental design prioritized minimizing false negatives as the primary clinically relevant error, no additional class-balancing intervention was applied to the final model. This choice is consistent with the observed tendency of the model to favor the pneumonia class.

Despite this reduced specificity, the model's sensitivity to pneumonia remained consistent with the strong pneumonia-class performance of the individual Xception and InceptionV3 models observed throughout this study (Section 4.2), missing only 4 of 390 actual pneumonia cases on the test set. This suggests that, in its current form, the model may be best suited as a pre-screening tool to flag likely pneumonia cases for radiologist review, prioritizing sensitivity over specificity.

5. Discussion

This study systematically analyzed the effects of various techniques that influence model performance when applying deep learning technology to pneumonia diagnosis using chest X-ray images, going beyond simply achieving high classification accuracy. A series of controlled experiments demonstrated that 0–1 pixel normalization significantly improved training stability and convergence, while Xception, InceptionV3, and DenseNet121 consistently achieved the highest classification performance among the six architectures compared. Furthermore, the results show that decision-level ensembling achieved more stable convergence and higher overall performance than feature-level fusion, which was

prone to overfitting due to its high-dimensional combined feature space. These findings provide practical insights for optimizing CNN-based pneumonia classification models under controlled experimental conditions. The significance of this study is as follows.

First, we empirically demonstrated that pixel value normalization is an essential pre-processing step for ensuring the models' training stability and convergence speed.

Second, through repeated comparisons across multiple independent runs, InceptionV3, DenseNet121, and Xception formed the top-performing group, showing closely spaced mean validation accuracies and substantially lower losses than the remaining architectures. InceptionV3 achieved the numerically highest mean accuracy, followed by DenseNet121 and Xception. Because no formal inferential statistical test was performed, these results are interpreted descriptively and do not establish statistical equivalence or significant pairwise differences among the three models. Xception and Inception V3, both grounded in parallel multi-scale filter processing, were carried forward to the subsequent optimization experiments, while DenseNet121—based on a structurally distinct feature-reuse design—was reserved for future work. Although Inception V3 and Xception are commonly associated with an input resolution of 299×299 pixels, both remained among the best-performing models under the typical 256×256 condition used in this study. This indicates that the standardized resolution allowed their architectural strengths to be observed within the controlled comparison. Nevertheless, a higher input resolution can affect the preservation of fine radiographic patterns and alter the absolute performance or magnitude of the differences among architectures; because a resolution-specific comparison was not conducted, this effect could not be quantified in the present study.

Third, we identified clear differences in ensemble methods. Feature-level fusion carries a high risk of overfitting due to the curse of dimensionality in high-dimensional feature spaces. Conversely, by combining the independent predictions of the individual models rather than their internal feature representations, decision-level fusion avoided this overfitting tendency and achieved both stable convergence and the highest validation accuracy and lowest validation loss among all models compared in this study.

Fourth, under the revised protocol, the test set was not used for performance-based model selection. All decisions regarding normalization, architecture, learning rate, and ensemble design were made using the new validation set. The test set was accessed during data preparation for overlap screening and was subsequently used to report the performance of the finalized model. Accordingly, the reported test results represent the final performance evaluation of the selected model under the revised protocol.

One dataset-related limitation should be considered when interpreting the results. As described in Section 3.1, pool images sharing a patient identifier or a near-duplicate match with the test set were identified and excluded prior to re-splitting to prevent leakage. However, because the normal class does not carry a patient identifier in the redistributed Kaggle dataset, patient-level overlap involving normal-class images could not be fully verified, either between the training/validation subsets and the test set or between the training and validation subsets themselves; only image-level (exact- and near-duplicate) leakage could be ruled out for this class. The strong validation-set performance for the normal class, contrasted with the marked drop in specificity observed on the fixed test set (Section 4.5), is consistent with residual patient-level overlap between the training and validation subsets for this class.

Because the design of this study prioritized minimizing false negatives rather than balancing performance across both classes, additional class-balancing and domain-adaptation strategies for the normal class were not investigated and remain important directions for future work.

In future research, we plan to extend the various techniques and models proposed in this study to diverse medical imaging datasets in order to verify their generalizability and validity. This will allow us to evaluate the adaptability and stability of the models based on data characteristics and derive structural improvements for more sophisticated performance enhancements. Ultimately, such follow-up studies may contribute to establishing more general optimization guidelines for deep learning-based diagnostic models after validation on multiple datasets and clinical settings.

The novelty of this study does not lie in proposing a new CNN architecture but rather in providing a systematic comparative framework that identifies the relative contribution of each major design factor under controlled experimental conditions. The findings suggest that, within the experimental setting of this study, optimizing the entire modeling pipeline may improve the robustness of CNN-based pneumonia classification models beyond architectural improvements alone.

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